

Amrita School of Engineering, Chennai

Course Delivery Plan

Course Code :Course Name	23MEE209: THEORY OF MECHANISMS AND MACHINES	Program	Robotics and Artificial Intelligence
L - T - P - C	3-0-3-4	Semester / Year	III/II
Name(s) of the Faculty	Dr. Shakti P Jena/Dr. Dev Singh Chauhan	Pre-requisite	Engineering Mechanics.
Course Mentor	Dr. Shakti P Jena	Academic Year	2026-27
Course Overview	This course will deal with kinematic analysis of mechanisms and machines. It will include motion and force transmission analysis of linkage mechanisms, cam and geared transmissions. The course will demonstrate various concepts by working out problems relevant to real-life applications of mechanisms. The course is expected to help students in their basic understanding and use of kinematic analysis.		

Course Objectives		Course Outcomes	
1	To familiarize with fundamental definitions and classification of mechanisms.	CO1	Understand the fundamentals of mechanisms to design mechanisms and machines.
2	To enable the students to perform kinematic synthesis and analysis of planar mechanisms with lower and higher pairs.	CO2	Perform kinematic analysis and synthesis of planar mechanisms.
3	To impart knowledge on dynamic analysis and balancing of masses.	CO3	Conduct a kinematic study of cams, gears and gear trains.
4	To familiarize with mathematical modelling and analysis of mechanical vibration systems.	CO4	Perform static force analysis of planar mechanisms and its application in robotics.
		CO5	Formulate the dynamic equilibrium equations of free longitudinal, transverse and torsional vibration systems and solve them

Syllabus

Unit 1

Basics of Mechanisms: Definitions - link, kinematic pair, kinematic chain, mechanism and machines. Degrees of freedom - mobility – Kutzbach criterion - Grashoff's law. Kinematic inversions -four-bar chain and slider-crank - mechanical advantage- transmission angle.

Kinematics Analysis of Mechanisms: Displacement, velocity and acceleration in simple planar mechanisms – Graphical method (Relative velocity method) - Coriolis component of acceleration.

Synthesis of Mechanisms: Types of synthesis - Three-position synthesis of planar mechanisms - Freudenstein equation.

Unit 2

Kinematics of Cams: Classification of cams and followers, nomenclature – Cams with different follower motions.

Kinematics of Gears and Gear Trains: Gears – terminology, fundamental law of gearing. Interference and undercutting.

Gear Trains – simple, compound and epicyclic gear trains.

Unit 3

Static force analysis – Analytical/graphical approach: Calculation of stalling and dynamic torque in robot manipulators; selection of Gear boxes/reducers in robot manipulators.

Vibrations: Basic concepts and types of vibration - single degree of freedom system. Free longitudinal vibrations: Equations of motion, undamped and damped free vibration of a single degree of freedom system, logarithmic decrement. Natural frequency of free transverse vibrations, critical speed of simple shafts. Natural frequency of free torsional vibrations.

Lab Components

- Identification of kinematics links, pairs, and chains in a mechanism & demonstration of various mechanisms using MAKIT - MASTER Mechanisms Kit and MAKIT - HITECH Controller Kit.
- Using ADAMS software, modelling and analysis of various planar mechanisms viz. slider-crank mechanism and its inversions, four-bar mechanism and its inversions, 6-bar chains, crank and slotted lever and Whitworth quick return mechanisms.
- Analysis of cam profiles
- Free vibration of spring-mass system
- Critical speed of shafts
- Bifilar, trifler and compound pendulums

Textbook(s)

T1. Uicker, John Joseph, Gordon R. Pennock, and Joseph Edward Shigley. Theory of machines and mechanisms. Vol. 1. New York, NY: Oxford University Press, 2011.

T2. Ambekar A.G., Mechanism and Machine Theory, Prentice Hall of India, New Delhi, 2007.

Reference(s)

R1. Norton, Robert L. Kinematics and dynamics of machinery. McGraw-Hill Higher Education, 2011.

R2. Rattan, Sarjit S. Theory of machines. Tata McGraw-Hill Education, 2014.

R3. Ghosh, Amitabha, and Asok K. Mallik. Theory of mechanisms and machines. Affiliated East-West Press Private Limited, 2002.

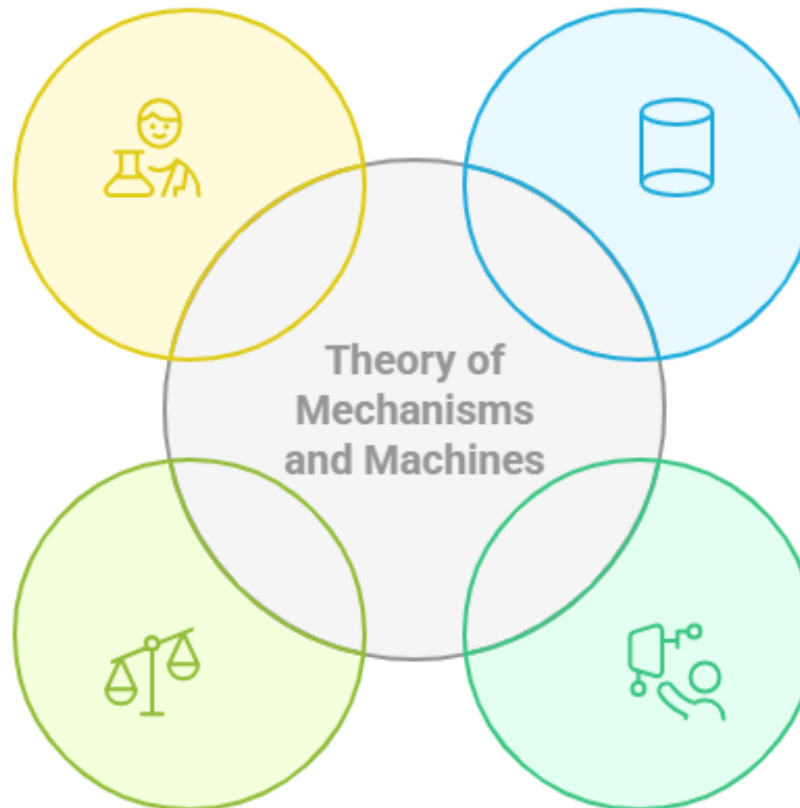
R4. David H. Myszka, Machines and Mechanisms: Applied Kinematic Analysis, 4th Edition, Prentice Hall, 2012.

Concept Map

Understanding Mechanisms and machines

Lab Components

Provides practical experience to reinforce theoretical knowledge.



Kinematics

Focuses on motion without considering forces.

Static Force Analysis

Examines forces in equilibrium within mechanisms.

Synthesis of Mechanisms

Involves designing mechanisms to achieve specific motions.

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Evaluation and Grading

Internal (60)				External (40)	Total
Components		Marks	Total Marks	End Semester = 40	Internal + External = 100
Mid Sem		30	30		
Continuous Assessment	Lab Worksheets	10	30		
	Lab mid exam	5			
	Lab end exam	5			
	Capstone Project	5			
	Class participation	5			

Programme Outcome (PO)

PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identity, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods, including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognise the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
PSO1	Design and develop cost-effective robotic systems catering to Industrial and Societal requirements.
PSO2	Develop cost-effective, safe, and efficient AI-based automation systems for manufacturing applications, focusing on product development and process improvement.
PSO3	Apply the acquired knowledge and skills in AI to address real-life multidisciplinary engineering problems.

CO – PO Affinity Map

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO															
CO 1	3	2	1		2				1	1		1	1		1
CO 2	3	3	2	2	3				2	1		1	1		1
CO 3	3	2	2		1				1	1		1	1		1
CO 4	3	3	2	1	2				1	1		1			1
CO 5	3	3	2	1					1	1		1	1		1

3-Strong, 2-Moderate, 1 -Weak

Threshold (%)	Target (%)	Percentage of Students		
		Level1	Level 2	Level 3
50%	55%	50%	60%	70%

				https://youtu.be/B5ZsYnyXBMY		
15,16	Relative velocity method.	Smart Board	Problem Based Learning	https://www.youtube.com/watch?v=AOkckB3m1sA	CO2	T1, R2, R3
17,18	Three-position synthesis of planar mechanisms	Smart Board	Problem Based Learning	https://www.youtube.com/watch?v=gu_mkMwqGBY&list=PL1dIdvYPZGigyGVViFobeDcrU399OaGTx	CO2	T1, R2, R3
19,20	Freudenstein equation	Smart Board	Problem Based Learning and Concept Mapping	https://www.youtube.com/watch?v=gu_mkMwqGBY&list=PL1dIdvYPZGigyGVViFobeDcrU399OaGTx	CO2	T1, R2, R3
UNIT 2						
21	Introduction to concepts of Cams	Smart Board	Problem Based Learning and Concept Mapping	https://www.youtube.com/watch?v=55tKVBVQDUY&t=991s	CO3	T1, R2, R3
25,26	Mechanism of different types of CAM Follower Motion	Smart Board	Problem Based Learning	https://www.youtube.com/watch?v=HsXWewecMLE	CO3	T1, R2, R3
27,28	Displacement diagrams and layouts of plate cam profiles. Numerical Problems.	Smart Board	Problem Based Learning and Concept Mapping	https://www.youtube.com/watch?v=ESRsd1TeOSg	CO3	T1, R2, R3
29,30	Tangent Cam with Reciprocating Roller Follower and Circular Arc Cam with Flat-Faced Follower	Smart Board	Problem Based Learning	https://www.youtube.com/watch?v=PTIP1LCoXek	CO3	T1, R2, R3
31	Gears: Classification of gears - gear tooth terminology	Smart Board	Problem Based Learning	https://www.youtube.com/watch?v=BjkyYZ93Fbs&list=PLbRMhDVUMngdCkMipemSKP_dCgZLLfOe8&index=38	CO3	T1, R2, R3

32	Fundamental law of toothed gearing and involute gearing	Smart Board	Problem Based Learning and Concept Mapping	https://www.youtube.com/watch?v=wCu9W9xNwtI	CO3	T1, R2, R3
33	Length of path of contact and contact ratio - interference and undercutting	Smart Board	Problem Based Learning and Concept Mapping	https://www.youtube.com/watch?v=yYAw79386WI	CO3	T1, R2, R3
34	Gear trains: Simple & compound gear trains, Differentials	Smart Board	Problem Based Learning and Concept Mapping	https://www.youtube.com/watch?v=lu_Qw4Y4XRQ&list=PLbRMhDVUMngdCkMipemSKP_dCgZLLfOe8&index=39	CO3	T1, R2, R3
35-36	Velocity ratio of epicyclic gear trains, Numerical Problem	Smart Board	Problem-Based Learning	https://www.youtube.com/watch?v=5f3wBIRY8dQ&list=PLbRMhDVUMngdCkMipemSKP_dCgZLLfOe8&index=40	CO3	T1, R2, R3
UNIT 3						
37-38	<i>Static force analysis</i> – Analytical/graphical approach: Calculation of stalling and dynamic torque in robot manipulators;	Smart Board	Problem-Based Learning	https://www.youtube.com/watch?v=Xicl4LQ9QZE&list=PLbRMhDVUMngdCkMipemSKP_dCgZLLfOe8&index=29 https://www.youtube.com/watch?v=wIGJAmYmzTU&list=PLbRMhDVUMngdCkMipemSKP_dCgZLLfOe8&index=35	CO4	T1, R2, R3
39	Selection of Gear boxes/reducers in robot manipulators.	Smart Board	Problem-Based Learning	https://www.youtube.com/watch?v=2PDsCJGXt68&list=PLbRMhDVUMngdCkMipemSKP_dCgZLLfOe8&index=30 https://www.youtube.com/watch?v=kZvZtbB4Ppo&list=PLbRMhDVUMngdCkMipemSKP_dCgZLLfOe8&index=31	CO4	T1, R2, R3

				hDVUMngdCkMipemSKP_dCgZLLfOe8&index=32		
40	Vibrations: Basic concepts and types of vibration - single degree of freedom system. Free longitudinal vibrations:	Smart Board	Problem-Based Learning	https://www.youtube.com/watch?v=vpDZyueADn8&list=PLp6ek2hDcoNBihD_JCzIvc-t_gApqMGKd&index=2 https://www.youtube.com/watch?v=LnnRY3BdCs4&list=PLp6ek2hDcoNBihD_JCzIvc-t_gApqMGKd&index=4	CO5	T1, R2, R3
41-42	Equations of motion, undamped and damped free vibration of a single degree of freedom system, logarithmic decrement.	Smart Board	Problem-Based Learning	https://www.youtube.com/watch?v=wbzVGwj9ZsM&list=PLp6ek2hDcoNBihD_JCzIvc-t_gApqMGKd&index=6	CO5	T1, R2, R3
43-44	Natural frequency of free transverse vibrations, critical speed of simple shafts.	Smart Board	Problem-Based Learning	https://youtu.be/ErN4c3B-Ve8	CO5	T1, R2, R3
45	Natural frequency of free torsional vibrations.	Smart Board	Problem-Based Learning	https://youtu.be/ErN4c3B-Ve8	CO5	T1, R2, R3

LAB SESSION

1,2,3	Bifilar, trifler		Problem-Based Learning	Lab Manual	CO5	T1, R2
4,5,6	Compound pendulums		Problem-Based Learning	Lab Manual	CO5	T1, R2
7,8,9	Identification of kinematics links, pairs, and chains in a mechanism	MAKIT kit	Problem-Based Learning	-	CO1	T1, R2
10,11,12	Demonstration of mechanism using the MAKIT kit	MAKIT kit	Problem-Based Learning	-	CO2	T1, R2
13,14,15	Demonstration of Tool bars and functions in ADAMS	ADAMS Student Version Software	Problem-Based Learning	Lab Manual	CO1	T1, R2
16,17,18	Four Bar Mechanism and its	MATLAB/ADAMS	Problem-Based	Lab Manual	CO2	T1, R2

	inversions	Student Version Software	Learning			
19,20,21	Slider Crank Mechanism and its inversions	MATLAB/ADAMS Student Version Software	Problem-Based Learning	Lab Manual	C02	T1, R2
22,23,24	LAB MID SEM EXAM					
25,26,27	Six bar Mechanism with Ternary Link	MATLAB/ADAMS Student Version Software	Problem-Based Learning	Lab Manual	CO2	T1, R2
28,29,30	Crank and slotted lever	MATLAB/ADAMS Student Version Software	Problem-Based Learning	Lab Manual	CO2	T1, R2
31,32,33	Quick Return Mechanism	MATLAB/ADAMS Student Version Software	Problem-Based Learning	Lab Manual	CO2	T1, R2
34,35,36	Analysis of cam profiles		Problem-Based Learning	Lab Manual	CO3	T1, R2
37,38,39	Free vibration of spring-mass system		Problem-Based Learning	Lab Manual	CO5	T1, R2
40,41,42	Critical speed of shafts		Problem-Based Learning	Lab Manual	CO5	T1, R2
43,44,45	LAB END SEM					

***Class Participation Requirement:**

- (i) All students must maintain a minimum of 75% attendance in the course. This requirement is crucial for academic success and eligibility to appear for mid-term and end-semester examinations.
- (ii) Students who fail to meet the 75% attendance threshold will not be permitted to appear for the mid-term and end-semester exams. Requests for exceptions will not be considered under any circumstances.
- (iii) Attendance will be calculated up to three days prior to the commencement of the mid-term and end-semester exams, as per the academic calendar.

Faculty

Course Mentor

Chairperson/ Program Head